

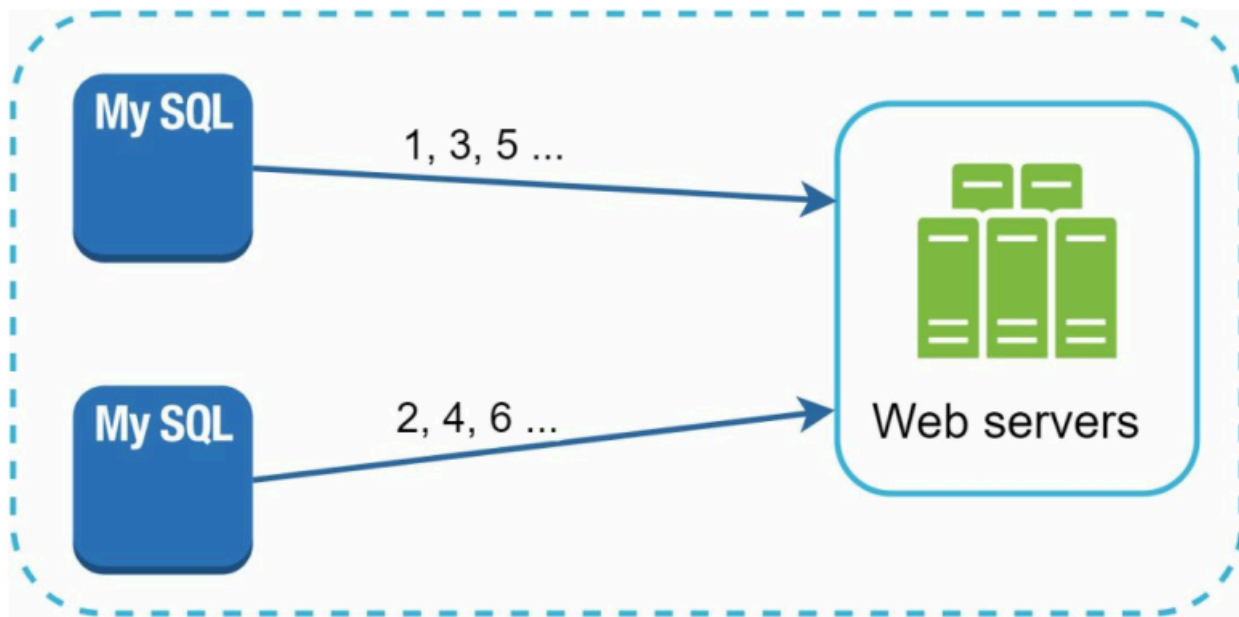
Design a Unique ID Generator - Step 2: Propose High-Level Design & Get Buy-In

1. Possible Approaches

Multiple approaches can be used to generate unique IDs in distributed systems:

- Multi-master replication
- Universally Unique Identifier (UUID)
- Ticket server
- Twitter Snowflake approach

2. Multi-Master Replication Approach



How It Works

- Uses database **auto_increment** feature
- Instead of incrementing by 1, increment by **k**
 - k = number of database servers

Example

- If $k = 2$:

None

Server 1 → 1, 3, 5, 7...

Server 2 → 2, 4, 6, 8...

Each server generates IDs independently

Advantage

- Improves scalability compared to single database

Drawbacks

- Hard to scale across **multiple data centers**
- IDs are **not globally time-ordered**
- Difficult to handle:
 - Adding new servers
 - Removing servers

3. Key Takeaway

Multi-master replication:

- Simple idea

- Solves some scalability issues
- But **fails in distributed, large-scale environments**